AI STRATEGY

AI PAIN POINTS

A Comprehensive Analysis of AI Implementation Challenges

45	42	11
Primary Insights	Secondary Insights	Pain Point Categories

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Executive Summary

This report presents a comprehensive synthesis of **87 AI pain points** gathered through primary research (interviews with AI practitioners) and secondary research (industry reports, academic literature and market analyses). Together, these two lenses reveal a nuanced picture of where AI initiatives succeed and, more importantly, why they fail.

> ***Core Finding: AI transformation is primarily an organisational and systems challenge, not a purely technical one. Both research streams converge on this conclusion independently.***

Six macro-categories concentrate the overwhelming majority of pain points:

- **Strategy and Organisational Alignment** — Vague objectives, misaligned expectations, leadership/team disconnect.
- **Data and Model Development** — Quality gaps, bias, annotation costs, overfitting, limited generalisability.
- **Infrastructure and Integration** — Compute constraints, cloud costs, legacy system compatibility, latency.
- **Operationalisation and MLOps** — Prototype-to-production failures, monitoring gaps, model drift.
- **Security, Privacy and Governance** — Prompt injection, agent hijacking, shadow AI, regulatory exposure.
- **Human Factors, Trust and Adoption** — Job-loss fears, overtrust, explainability gaps, change resistance.

A particularly striking insight is the **divergence between the two research streams**: secondary sources reflect the *classic machine-learning lifecycle* (data, models, infrastructure), while primary interviews surface a new wave of *generative AI and agentic system risks* — security vulnerabilities, governance gaps, and human misuse — that traditional frameworks have yet to fully address.

45	42	11	8
Primary Insights	Secondary Insights	Categories	Shared Themes

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Methodology

The analysis combines two complementary research streams designed to capture both ground-level practitioner experience and broad industry consensus.

Dimension	Primary Research	Secondary Research
Source	Direct interviews and practitioner observations	Industry reports, academic studies, market analyses
Volume	45 pain points	42 pain points
Focus	Real-world GenAI and agentic deployment	Systemic ML and data science challenges
Perspective	Operational and emerging risks	Structural and established risks
Time horizon	2023–2026 (generative AI era)	2018–2026 (full ML lifecycle)

Pain points from both streams were independently coded into **11 thematic categories** using open-card sorting. Categories were then merged where labels overlapped, yielding a unified taxonomy enabling direct cross-stream comparison.

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Section 1 — Pain Point Landscape Overview

Across 87 collected pain points, the research surfaces a rich and multi-dimensional picture of AI implementation friction. The chart below compares the volume of pain points per category across both research streams.

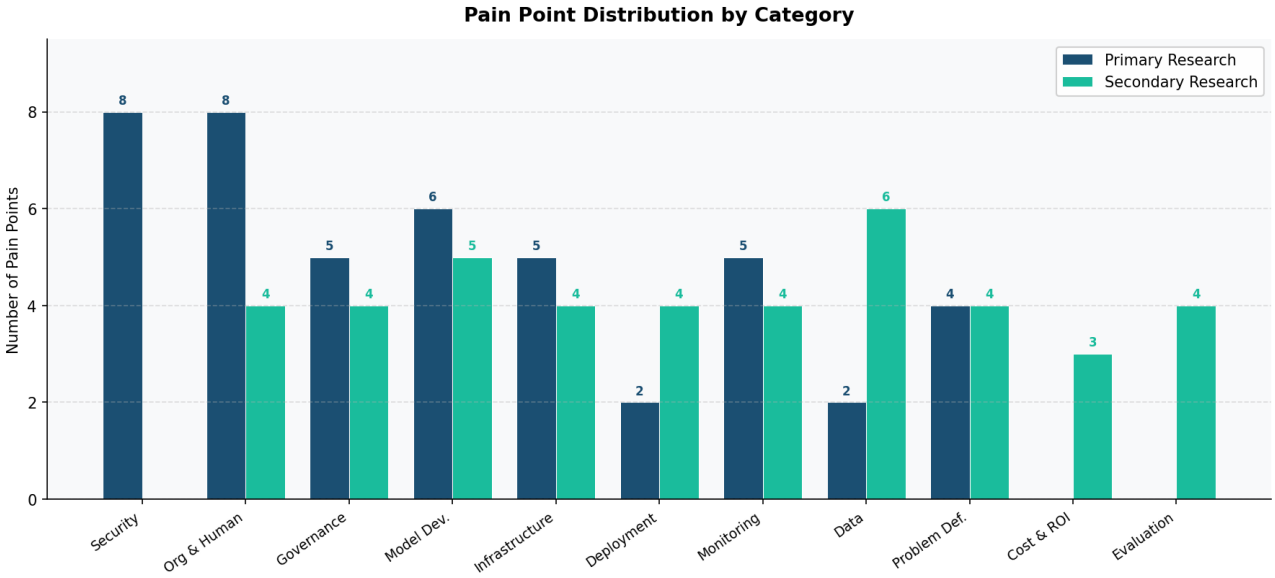

Figure 1 — Pain point count by category: Primary (blue) vs. Secondary (teal) research.

Research Stream Breakdown

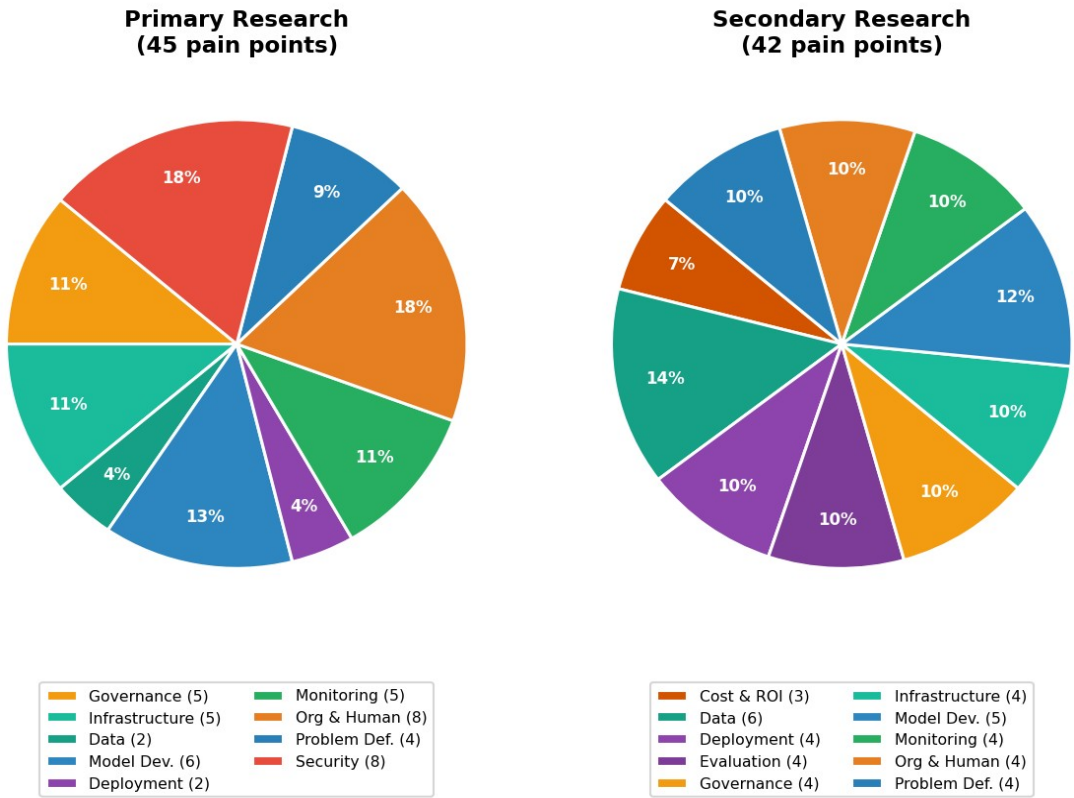

Figure 2 — Distribution of pain points by category for each research stream.

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Section 2 — Category Deep Dive

Security Vulnerabilities

Primary: **8** | Secondary: **1**

The single sharpest divergence between the two research streams. Primary research identifies **8 security pain points** — all specific to generative AI and agentic architectures — compared to just 1 generic mention in secondary sources. This gap reflects the lag between security risk discovery and real-world exploitation. Organisations should treat this as a leading indicator of operational problems to come.

Pain Point	Description
Prompt injection attacks	Attackers manipulating prompts to force AI agents to execute unintended actions.
Agent goal hijacking	Malicious inputs redirecting autonomous agents to perform unintended tasks.
Remote code execution	AI development environments enabling attackers to execute arbitrary code.
Supply chain vulnerabilities	Malicious packages or plugins compromising AI development environments.
Data exfiltration via AI agents	Agents accessing and leaking sensitive data through integrations or tools.
Agent privilege abuse	Agents exploiting excessive permissions to access unauthorized systems or data.
Rogue autonomous agents	Agents performing unintended or harmful actions without human oversight.
Human-agent trust exploitation	Users trusting AI outputs excessively, enabling exploitation or unnoticed errors.

Governance, Risk and Compliance

Primary: **5** | Secondary: **4**

Primary research expands significantly on the governance theme, identifying systemic institutional gaps rather than project-level compliance issues. The five primary pain points reflect the growing regulatory and governance burden that will reshape AI deployment strategies across all industries in the near term.

Pain Point	Description
Shadow AI adoption	Employees or teams using AI tools without governance or security oversight.
Lack of AI governance frameworks	Organisations deploying AI without formal governance policies or oversight mechanisms.
Regulatory uncertainty	Industrial or enterprise AI projects slowed by unclear or evolving regulatory requirements.
Certification and compliance burden	AI solutions needing extensive documentation and traceability to meet industry certification requirements.
GDPR and privacy concerns	Risks related to compliance with privacy regulations when processing sensitive data with AI.

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Organizational and Human Factors

Primary: **8** | Secondary: **4**

Primary research surfaces a richer taxonomy of organisational challenges than secondary research, identifying eight distinct dimensions. The key difference is emphasis: primary research highlights structural and cultural maturity gaps at the organisational level, including AI literacy among decision-makers, trust dynamics, and misalignment between executive ambition and operational capacity.

Pain Point	Description
Lack of AI literacy	Decision-makers lacking understanding of AI capabilities and limitations.
Skill misperception	People assuming basic library knowledge is sufficient for AI expertise.
Low organisational AI maturity	Companies unsure where to start or how to structure AI initiatives.
Data management immaturity	Organisations lacking proper internal processes to organise and manage data.
Client trust barriers	Smaller companies struggling to gain trust when offering AI solutions compared to large enterprises.
Resistance to AI adoption	Fear among employees that AI may replace their jobs leading to reluctance in adoption.
Fear of AI errors	Organisations concerned about deploying AI systems that may produce incorrect outputs.
Leadership/team misalignment	Executives pushing AI adoption while operational teams resist or lack ownership.

Model Development Issues

Primary: **6** | Secondary: **5**

Primary research adds a distinct dimension to model development challenges: the specific failure modes of generative AI systems. Unlike classical ML issues (overfitting, reproducibility), these problems arise from the nature of large language models and multimodal generation systems.

Pain Point	Description
Prompt interpretation errors	AI systems frequently misinterpreting user instructions or prompts.
Iteration-heavy workflows	Users needing many prompt iterations to achieve acceptable results.
Avatar realism limitations	Generated avatars showing unnatural facial expressions or robotic behaviours.
Temporal inconsistency in video	Objects or characters changing appearance across frames in AI-generated video.
Physics inconsistency	Generated video motion sometimes defying realistic physical behaviour.
AI-generated code reliability	Generated code often containing errors that the AI struggles to correct.

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Infrastructure and Tooling

Primary: **5** | Secondary: **4**

Primary research infrastructure pain points reflect a shift toward edge and embedded AI deployment, with compute and energy constraints becoming increasingly prominent as AI moves beyond cloud-centric architectures. The cost dimension also takes a different form in primary research: the credit-based pricing of generative AI tools.

Pain Point	Description
High compute and energy requirements	Large AI models require significant compute resources and energy, especially in robotics contexts.
Edge deployment limitations	Difficulty embedding large generative models directly on devices due to hardware limitations.
Cloud dependency architectures	Reliance on hybrid cloud architectures to compensate for limited on-device processing power.
Credit-based cost escalation	Iterative prompting and experimentation consuming credits and increasing costs significantly.
Scaling infrastructure complexity	AI systems requiring significant infrastructure expertise to scale to production workloads.

Monitoring, Maintenance and Lifecycle

Primary: **5** | Secondary: **4**

Primary research introduces monitoring challenges specific to generative AI and no-code AI development platforms, including platform dependency risk and the technical debt accumulation that characterises AI-generated code.

Pain Point	Description
Technical debt from AI-generated code	Automatically generated code lacking structure or documentation leading to long-term maintenance issues.
Platform dependency risk	Loss of code access or functionality if AI development platforms discontinue services.
Backup and continuity risks	No-code platforms sometimes lacking mechanisms to export or back up generated applications.
Lack of observability	Limited monitoring tools to track AI outputs and behaviours in production.
AI error detection challenges	Difficulty identifying when AI outputs are incorrect or hallucinated.

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Problem Definition and Business Alignment

Primary: **4** | Secondary: **4**

Primary research provides a more market-facing view of problem definition challenges, including the economic disruption AI creates for existing business models and the systemic gap between AI marketing claims and technical reality.

Pain Point	Description
AI marketing vs real capability	Overhyped expectations created by AI marketing narratives.
Overestimation of autonomous agents	Belief that agents operate fully autonomously without supervision or guardrails.
Integrating probabilistic systems	Organisations struggling to integrate probabilistic AI systems into deterministic workflows.
Business model disruption	AI reducing time-based consulting work and forcing shifts to value-based or subscription pricing models.

Data-Related Challenges

Primary: **2** | Secondary: **6**

Data challenges represent the single largest operational bottleneck in secondary research. Unlike model or infrastructure problems, data issues are uniquely difficult to solve because they require cross-functional ownership and often reveal organisational dysfunctions. Primary research surfaces generative-AI-specific data problems: context window limitations and statistical hallucinations.

Pain Point	Description
Context window limitations	AI systems forgetting earlier information when processing large files or datasets.
Statistical hallucinations	AI generating incorrect statistical conclusions when analysing large datasets.

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Section 3 — Emerging Generative AI Pain Points

Primary research reveals a distinct cluster of pain points that are unique to — or dramatically amplified by — the adoption of large language models, generative media tools, and agentic AI systems.

Prompting Inefficiency

Users require many iterations to achieve acceptable outputs, generating productivity drag and unpredictable costs. Token-based pricing amplifies this effect.

Output Quality and Hallucinations

LLMs generate plausible but factually incorrect content. Statistical reasoning errors are particularly dangerous in analytical contexts. Visual generation tools suffer from temporal inconsistency and physics violations.

Code Generation Reliability

AI-generated code contains hidden errors that the model struggles to self-diagnose, creating technical debt. No-code platforms lack adequate debugging mechanisms, increasing dependency on vendor support.

Agentic System Risks

Autonomous agents operating with broad permissions introduce novel risks: goal hijacking, privilege escalation, and the potential for harmful autonomous actions without adequate human oversight.

Context Window Constraints

Current context window limitations cause AI systems to lose earlier information in long sessions reducing reliability in document processing and multi-turn workflows.

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Section 4 — Primary vs Secondary Research Comparison

Coverage Radar

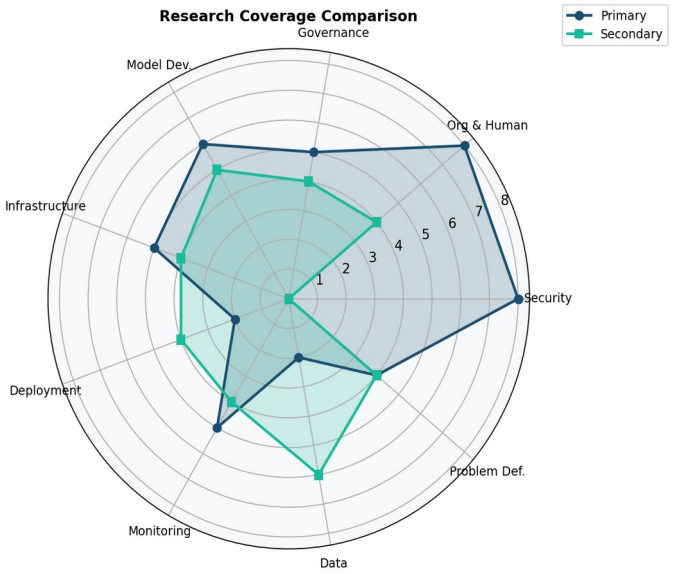

Figure 3 — Radar chart comparing research coverage across shared categories.

Shared Category Intensity

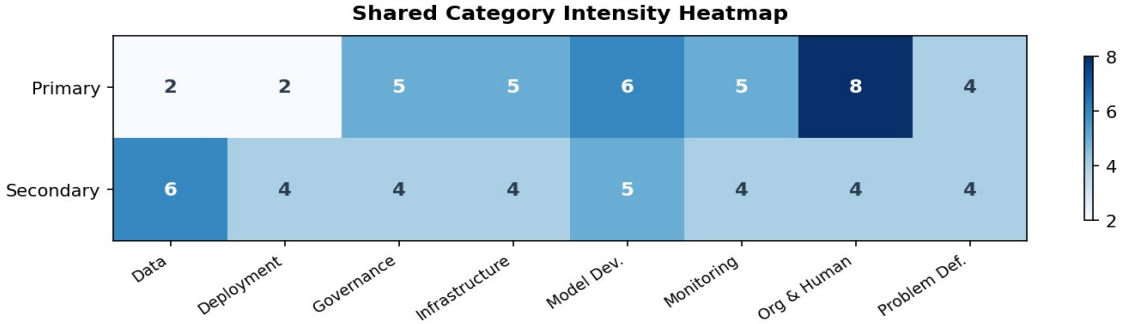

Figure 4 — Heatmap of pain point counts for categories present in both research streams.

Convergence: Where Both Sources Agree

Despite their different perspectives, both research streams converge strongly on five domains, suggesting these represent the most robust and universally recognised AI pain points:

✓ **Organisational and Human Barriers**
Both datasets identify talent shortages, cross-functional communication failures, weak AI literacy in leadership, and cultural resistance as primary adoption obstacles. The volume of organisational issues (4 in secondary, 8 in primary) confirms that human factors outweigh technical factors in determining AI success.

✓ **Model Development Complexity**
Hyperparameter tuning overhead, architecture selection difficulty, reproducibility failures, and the marginal value of incremental improvements appear in both sources. Even the most technically sophisticated organisations struggle with making model development efficient and cost-effective.

✓ **Infrastructure and Tooling Fragmentation**
Both datasets highlight fragmented MLOps ecosystems, compute constraints, and environment reproducibility as persistent challenges. This consensus across practitioner and analyst perspectives confirms that AI infrastructure remains meaningfully less mature than traditional software engineering infrastructure.

✓ **Problem Definition and Business Alignment**
Solving the wrong problem, unclear use cases, misaligned expectations, and the pursuit of AI as a technology goal rather than a business outcome are flagged by both sources. This convergence identifies problem framing as a high-leverage intervention point that is systematically underinvested.

✓ **Monitoring and Lifecycle Management**
Model drift, retraining complexity, and maintenance burden appear in both research streams, confirming that the AI lifecycle extends far beyond initial deployment and requires continuous investment that is rarely budgeted for.

Divergence: What Each Source Uniquely Emphasises

A systematic comparison reveals not only what challenges AI presents, but how these challenges are perceived differently depending on one's vantage point. Understanding this divergence is strategically important because it predicts which problems are being actively managed and which are being overlooked.

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Primary Research — Exclusive Emphasis

→ AI Security Vulnerabilities (8 issues, 0 in secondary)

The entire security category is absent from secondary research. Prompt injection, agent goal hijacking, data exfiltration via agents, supply chain vulnerabilities, and rogue autonomous agents represent a new attack surface that practitioners have not yet systematically encountered. Primary research correctly identifies these as the next frontier of AI operational risk.

→ Governance and Regulatory Frameworks (5 issues vs. 4)

Primary research goes beyond individual project compliance to address institutional governance gaps: shadow AI adoption, missing governance frameworks, GDPR compliance at scale, and regulatory uncertainty. These are enterprise-wide challenges that transcend individual project teams.

→ Generative AI-Specific Failures

Primary research captures failure modes unique to large generative models — prompt interpretation errors, hallucinations, avatar realism limitations, temporal inconsistency in video generation — that secondary research does not yet systematically document, likely because generative AI is a more recent deployment paradigm.

Secondary Research — Exclusive Emphasis

→ Data Pipeline Reality (6 issues vs. 2 in primary)

Secondary research's 6 data issues versus 2 in primary research reflects the researcher's daily encounter with missing labels, inconsistent formats, siloed databases, and the sheer cost of data preparation. Practitioners often treat data as a solved prerequisite; researchers know it is a permanent construction project.

→ Deployment Friction (4 issues vs. 2 in primary)

Productionisation gaps, latency constraints, and integration complexity appear far more prominently in secondary research. The 'last mile' of AI deployment is a practitioner problem — researchers rarely follow models through to production.

→ Economic Reality — Cost and ROI (3 issues, effectively 0 in primary)

Cost and ROI challenges and evaluation metrics misalignment reflect practitioner accountability for business outcomes that academic and policy researchers do not face directly.

Strategic Note: The divergence between what practitioners report and what researchers identify is not a discrepancy to resolve — it is a signal to act on.

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Key Divergences at a Glance

The comparative analysis reveals a meaningful structural shift in the nature of AI challenges over time:

Dimension	Primary Research (generative AI era)	Secondary Research (classical ML era)
Dominant challenge	Security and governance	Data quality and quantity
Technical focus	Output reliability and hallucinations	Model performance and generalisation
Infrastructure concern	Edge deployment and cloud dependency	Compute cost and scalability
Organisational risk	Shadow AI and overtrust	Skill gaps and misaligned metrics
Lifecycle risk	Technical debt and platform dependency	Model drift and retraining
Security posture	Agent hijacking and prompt injection	Adversarial attacks (model-level)

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Section 5 — Cross-Cutting Insights and The AI Stack

When the full dataset of 87 pain points is analysed not by category but by the structural patterns they reveal, four systemic insights emerge that cut across all dimensions of the research.

5.1 The AI Problem Stack

The pain points can be organised into a layered stack that mirrors the AI development lifecycle. The critical insight from this structure is directional: **the lower the layer, the higher the frequency and impact of failures.** Yet organisations typically invest their attention and budget at the upper layers — model development and deployment — while the foundations below remain weak.

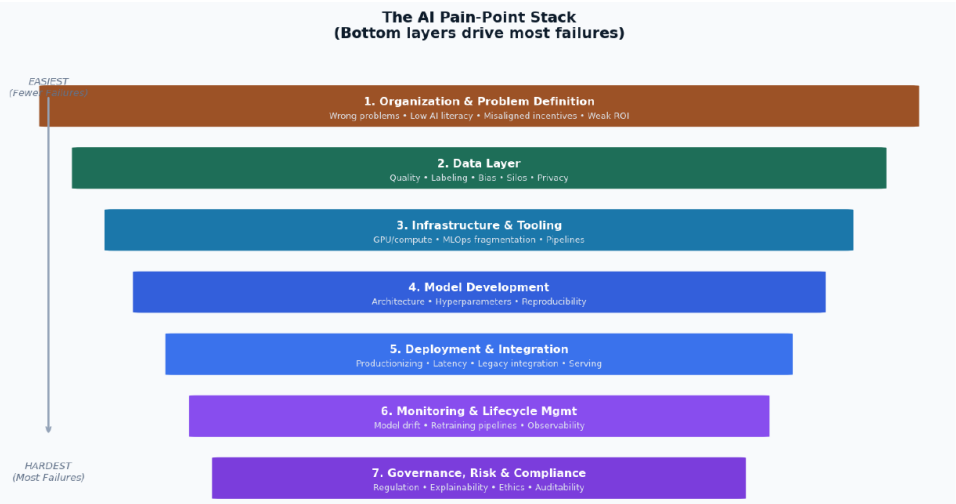

Figure 5 — The AI Pain-Point Stack. Width reflects relative prevalence of issues in that layer. Most failures originate at the organisational, data, and infrastructure layers, not at the model or deployment layers.

5.2 The AI Inversion Problem

A central paradox revealed by the combined dataset is that organisations direct their attention and investment toward the layers where problems are most visible and easiest to frame — model performance, algorithm selection — while systematically underinvesting in the layers where problems are most consequential: data pipelines, organisational alignment, problem definition. This 'AI Inversion' explains why AI ROI consistently disappoints despite rapid algorithmic progress.

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Layer	Where Organisations Focus	Where Problems Actually Live	Misalignment
Model	High (primary investment)	Medium (5 issues in secondary)	**Overinvested**
Data	Medium (reactive)	Very High (6 secondary, 2 primary)	**Underinvested**
Infrastructure	Low (assumed)	High (4 secondary, 5 primary)	**Underinvested**
Organisation	Very Low (HR problem)	Very High (4 secondary, 8 primary)	**Severely Underinvested**
Governance and Security	Low (compliance checkbox)	High (4 secondary, 13 primary)	**Critically Underinvested**

5.3 AI Is a Continuous System, Not a Deliverable

A third systemic insight concerns the nature of AI systems over time. Traditional software is built once and maintained reactively. AI systems, by contrast, are sensitive to change: user behaviour shifts, data distributions evolve, and model performance degrades continuously. The research documents this through monitoring and lifecycle pain points in both datasets, and through the economic pain points of ongoing operational costs.

Organisations that treat AI deployment as a project completion milestone systematically underinvest in the operational capabilities — retraining pipelines, monitoring infrastructure, drift detection, version management — required to sustain AI value.

5.4 The Three-Layer Problem Hierarchy

The combined dataset reveals that AI challenges exist simultaneously at three distinct levels that require different types of intervention:

Operational Layer (Secondary Research Dominant)

Data pipelines, model deployment, evaluation, and cost management. These are immediate, tangible, and typically addressed within project teams. They respond to technical and process improvements.

Engineering Layer (Both Research Sources)

Infrastructure, model development, monitoring, and lifecycle management. These are platform-level challenges that require organisation-wide standards, shared tooling, and MLOps maturity. They respond to platform investment and standardisation.

Risk and Governance Layer (Primary Research Dominant)

Security vulnerabilities, regulatory compliance, ethics, and shadow AI adoption. These are institutional challenges that require governance frameworks, legal engagement, and executive commitment. They respond to policy, oversight, and organisational accountability.

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Section 6 — Strategic Implications

01

Anchor AI in Business Problems, Not Technology

The highest leverage intervention in any AI programme is rigorous problem definition. Before any data collection, modelling, or infrastructure investment, organisations must establish clear business outcomes, measurable success criteria, and explicit decision flows that AI will support. The research documents four specific ways this fails: vague use cases, misaligned metrics, over-automation expectations, and scope creep.

02

Treat Data as a Strategic Asset, Not a Technical Input

With 8 data-related issues across both datasets, data management is the single most documented challenge in the research. Organisations must invest in data governance, labelling infrastructure, bias detection, and cross-functional data ownership as strategic capabilities — not IT projects. The organisations that consistently outperform in AI are those with mature data cultures, not those with the most sophisticated models.

03

Build MLOps as Organisational Infrastructure

Toolchain fragmentation, environment inconsistencies, and compute management challenges reflect the immaturity of AI infrastructure as a discipline. Organisations should standardise on an MLOps platform that covers the full lifecycle — data ingestion, feature engineering, model training, registry, serving, and monitoring — rather than assembling point solutions that create compounding complexity.

04

Anticipate the Security and Governance Wave

The 8 security vulnerabilities and 9 governance issues documented exclusively or predominantly in primary research represent the next wave of operational challenges. Prompt injection, agent hijacking, shadow AI, and GDPR compliance are not hypothetical risks — they will become practitioner problems within the near-term planning horizon. Organisations that build governance frameworks and AI-specific security protocols now will avoid reactive compliance scrambles later.

05

Invest in Organisational AI Readiness

The largest category in the combined dataset is organisational and human factors. No technical investment can compensate for skill gaps, cross-functional friction, unclear ownership, or cultural resistance to AI-driven decisions. Organisations should invest in AI literacy programmes for leadership, define explicit ownership structures for model performance and governance, and design cross-functional operating models that bridge business, data science, and engineering teams.

06

Budget for the Full AI Lifecycle

ROI misalignment, unclear timelines, and underestimated total cost of ownership reflect a systematic failure to account for the continuous nature of AI systems. Budgeting should include ongoing data labelling, model retraining, monitoring infrastructure, governance processes, and talent retention — not just initial development. The 'build and done' mental model consistently leads to abandoned AI systems.

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Conclusion and Key Takeaways

The convergent finding across all 87 documented pain points, across two distinct research methodologies, across eleven analytical categories, is this: **the primary constraint on AI value creation is not the sophistication of the algorithms — it is the maturity of the organisations, data ecosystems, and operational processes that surround them.**

1. AI failures are systemic, not algorithmic

Across both research streams, the root causes of AI project failure cluster around poor problem definition, weak data infrastructure, deployment complexity, and organisational misalignment — not algorithmic limitations. The era of 'AI can't do this yet' has largely given way to 'we didn't set this up to succeed.'

2. Generative AI has created an entirely new risk category

Security vulnerabilities specific to LLMs and agentic systems — prompt injection, agent hijacking, data exfiltration — represent a qualitative shift in the threat landscape. These risks are invisible to classical ML governance frameworks and demand immediate, dedicated attention from security and compliance teams.

3. The prototype-to-production chasm persists

Despite years of awareness, the operationalisation gap remains one of the most consistent pain points across both research streams. AI value is destroyed between the notebook and the production environment. MLOps maturity is a competitive differentiator.

4. Data remains the ultimate constraint

Secondary research — grounded in years of evidence — confirms that data quality, labelling, and silos continue to be the primary bottleneck for AI success. Generative AI compounds this through context window limitations and hallucination sensitivity to data quality.

5. The window for governance is now

The speed of AI adoption has outpaced the development of governance structures in most organisations. Shadow AI usage is widespread, policies are absent, and accountability is unclear. Organisations that establish governance frameworks now will be better positioned to scale AI responsibly and avoid costly regulatory or reputational incidents.

6. AI strategy is infrastructure strategy

Organisations that lead in AI adoption are not those with the largest research budgets or the most advanced models. They are those that have solved the unglamorous problems: clean data pipelines, clear use cases, cross-functional operating models, robust MLOps infrastructure, and governance frameworks that anticipate the risks primary research has already mapped.

> ***AI strategy is infrastructure strategy. Organisations that treat it as model strategy will systematically underperform those that invest in the organisational and technical foundations that make AI sustainable.***

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Glossary of Technical Terms

This glossary defines technical and specialist terms used throughout the report. It is intended to support readers who may be less familiar with AI and machine learning terminology.

A/B Testing	A controlled experiment in which two versions of a system (A and B) are compared by exposing different user groups to each. In AI, it is used to measure whether a new model actually improves real-world outcomes relative to the existing one.
Adversarial Attack	A deliberate attempt to fool an AI model by feeding it subtly manipulated inputs that cause it to make incorrect predictions. For example, adding imperceptible noise to an image can cause an image classifier to misidentify it with high confidence.
Agentic AI / Autonomous Agent	An AI system capable of taking sequences of actions, making decisions, and pursuing goals with minimal human intervention. Unlike a simple chatbot that responds to a single query, an agent can plan, use tools, browse the web, write code, or call external services on its own.
Agent Goal Hijacking	A security attack in which a malicious input redirects an autonomous AI agent away from its intended objective toward a different, harmful goal chosen by the attacker. Similar in concept to social engineering, but targeting the AI system rather than a human.
Agent Privilege Abuse	A situation where an AI agent exploits permissions it was legitimately granted — such as access to files, databases, or external APIs — to perform actions beyond its intended scope, whether through malicious manipulation or poor design.
Avatar Realism Limitations	The inability of current AI video and image generation tools to produce virtual human characters (avatars) with fully natural facial expressions, micro-movements, and behaviours. Generated avatars often appear robotic or uncanny to human observers.
Context Window	The maximum amount of text or data that a large language model can process in a single interaction. Content outside this window is effectively forgotten by the model. A limited context window becomes a problem when analysing long documents, codebases, or multi-session conversations.
Data Drift / Model Drift	The degradation in model performance that occurs when the real-world data the model encounters after deployment differs from the data it was trained on. For example, a fraud detection model trained on pre-2020 transaction patterns may become less accurate as payment behaviours evolve.
Data Exfiltration	The unauthorised transfer of data out of an organisation. In the context of AI agents, this refers to the risk that an agent with access to sensitive internal systems could be manipulated into leaking that data to an external party.
Data Leakage	A technical error in model training where information from the test or validation set inadvertently influences the training process, resulting in artificially inflated performance metrics that do not reflect real-world accuracy.

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Data Silos	Isolated data stores held by different teams, departments, or systems within an organisation that are not accessible or shared across the business. Silos are a major obstacle to AI because models typically require integrated, comprehensive datasets to perform well.
GDPR (General Data Protection Regulation)	A European Union regulation governing the collection, storage, processing, and use of personal data. It imposes strict requirements on organisations using AI systems that process personal information, including the right to explanation for automated decisions.
Hallucination	A phenomenon in which a large language model generates text that is fluent and confident but factually incorrect or entirely fabricated. For example, a model may invent a plausible-sounding court case, research paper, or statistic that does not exist.
Hyperparameter	A configuration setting that controls how a machine learning model is trained, as opposed to the parameters the model learns from data. Examples include learning rate, number of layers in a neural network, and batch size. Choosing the right hyperparameters requires significant experimentation.
Latency	The time delay between submitting a request to an AI system and receiving its response. High latency is a deployment problem when AI must operate in real-time contexts such as customer service, fraud detection, or robotics.
Large Language Model (LLM)	A type of AI model trained on vast quantities of text data to understand and generate human language. Examples include GPT-4, Claude, and Llama. LLMs power most modern chatbots, coding assistants, and document analysis tools.
MLOps (Machine Learning Operations)	The set of practices, tools, and infrastructure for deploying, monitoring, and maintaining machine learning models in production. MLOps bridges the gap between data science experimentation and reliable, scalable software deployment — analogous to DevOps for traditional software.
Observability	The ability to understand what is happening inside an AI system in production by monitoring its inputs, outputs, and internal states. Poor observability means errors, biases, and performance degradations go undetected until they cause visible harm.
Overfitting	A modelling error where a model learns the training data too precisely, including its noise and quirks, and therefore performs poorly on new, unseen data. An overfit model has memorised rather than generalised.
Physics Inconsistency	A failure mode in AI-generated video where objects or movements violate the laws of physics — for example, liquids flowing upward, objects passing through each other, or bodies moving in anatomically impossible ways.
Probabilistic System	A system that produces outputs based on statistical likelihoods rather than deterministic rules. AI models are inherently probabilistic — they provide the most likely answer given the input, which means the same input can occasionally produce different outputs. This clashes with traditional business software that always produces the same result given the same input.
Productionisation	The process of taking a machine learning model from a research or prototype environment and deploying it as a reliable, scalable component of a live system. Often more technically challenging than building the model itself.

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Prompt Fidelity	The degree to which an AI system accurately follows the intent of a user's instruction (prompt). Low prompt fidelity means the model frequently misinterprets, ignores, or partially fulfils what was asked, requiring many iterative corrections.
Prompt Injection	A security attack in which malicious instructions are embedded in content that an AI system reads — such as a document, email, or webpage — causing the model to execute those instructions instead of its intended task. Analogous to SQL injection attacks in traditional software.
Reproducibility	The ability to re-run a machine learning experiment and obtain the same result. Poor reproducibility — caused by random seeds, undocumented pipeline steps, or changing data — makes it difficult to validate, audit, or build on previous work.
ROI (Return on Investment)	A measure of the financial benefit generated by an investment relative to its cost. In AI projects, ROI is frequently difficult to measure because the benefits — improved decisions, productivity gains, risk reduction — are often indirect, delayed, or hard to attribute to the AI system specifically.
Rogue Autonomous Agent	An AI agent that performs actions outside its intended scope, either through manipulation by an attacker or due to flawed goal specification. Unlike a simple software bug, a rogue agent may actively pursue an unintended objective across multiple steps before human oversight can intervene.
Shadow AI	The use of AI tools by employees or teams within an organisation without the knowledge, approval, or oversight of IT, security, or leadership. Examples include staff using consumer AI chatbots to process confidential documents, or teams deploying unofficial AI tools outside approved procurement. Shadow AI creates security, compliance, and governance risks that the organisation is unaware of.
Supply Chain Vulnerability	A security weakness that enters a system through a third-party component, library, or plugin rather than the organisation's own code. In AI, this includes malicious packages in open-source ML libraries, compromised model weights, or infected plugins in AI development environments.
Technical Debt	The accumulated cost of shortcuts, quick fixes, and poorly structured code that must eventually be corrected. AI-generated code can accelerate technical debt accumulation because it often lacks documentation, follows inconsistent patterns, and is difficult to audit or maintain.
Temporal Inconsistency	A visual artefact in AI-generated video where an object, character, or scene element changes appearance, colour, or shape between frames in an unintended way, breaking the illusion of a coherent, continuous scene.
Underfitting	The opposite of overfitting: a model that is too simple to capture the underlying patterns in the data, resulting in poor performance on both training and new data. An underfit model has not learned enough from the available information.

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Annex A — Full Primary Research Pain Points (45)

The following table lists all 45 pain points collected from primary research (interviews and direct practitioner observations), grouped by thematic category.

Data-Related Challenges		
1	Context window limitations	
2	Statistical hallucinations in analysis	
Deployment and Integration Challenges		
1	Short output duration limits	
2	Prompt fidelity problems	
Governance Risk and Compliance		
1	Shadow AI adoption	
2	Lack of AI governance frameworks	
3	Regulatory uncertainty	
4	Certification and compliance burden	
5	GDPR and privacy concerns	
Infrastructure and Tooling Constraints		
1	High computational and energy requirements	
2	Edge deployment limitations	
3	Cloud dependency architectures	
4	Credit-based cost escalation	
5	Scaling infrastructure complexity	
Model Development Issues		
1	Prompt interpretation errors	
2	Iteration-heavy workflows	
3	Avatar realism limitations	
4	Temporal inconsistency in video generation	
5	Physics inconsistency in generated media	
6	AI-generated code reliability issues	
Monitoring Maintenance and Lifecycle Management		
1	Technical debt from AI-generated code	
2	Platform dependency risk	
3	Backup and continuity risks	
4	Lack of observability in AI systems	
5	AI error detection challenges	

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Organizational and Human Factors		
1	Lack of AI literacy in organizations	
2	Skill misperception	
3	Low organizational AI maturity	
4	Data management immaturity	
5	Client trust barriers	
6	Resistance to AI adoption	
7	Fear of AI errors	
8	Internal misalignment between leadership and teams	

Problem Definition and Business Alignment		
1	Misalignment between AI marketing and real capability	
2	Overestimation of autonomous agents	
3	Difficulty integrating probabilistic systems	
4	Business model disruption	

Security Vulnerabilities		
1	Prompt injection attacks	
2	Agent goal hijacking	
3	Remote code execution vulnerabilities	
4	Supply chain vulnerabilities in AI tools	
5	Data exfiltration via AI agents	
6	Agent privilege abuse	
7	Rogue autonomous agents	
8	Human-agent trust exploitation	

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Annex B — Full Secondary Research Pain Points (42)

The following table lists all 42 pain points identified through secondary research (industry reports, academic studies, and AI implementation analyses), grouped by thematic category.

Cost and ROI Challenges		
1	Diminishing returns	
2	Unclear ROI timelines	
3	Underestimated total cost of ownership	
Data-Related Challenges		
1	Data bias	
2	Data leakage	
3	Data silos	
4	Insufficient data quantity	
5	Lack of labeled data	
6	Poor data quality	
Deployment and Integration Challenges		
1	Integration complexity	
2	Latency and throughput issues	
3	Productionization gaps	
4	Scalability limitations	
Evaluation and Success Measurement		
1	Inappropriate evaluation metrics	
2	Limited A/B testing capability	
3	Offline-online mismatch	
4	Short-term optimization	
Governance Risk and Compliance		
1	Ethical concerns	
2	Insufficient auditability	
3	Regulatory non-compliance	
4	Security vulnerabilities	
Infrastructure and Tooling Constraints		
1	Environment inconsistencies	
2	High infrastructure costs	
3	Inadequate compute resources	
4	Toolchain complexity	

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Model Development Issues		
1	Hyperparameter sensitivity	
2	Inappropriate model selection	
3	Limited explainability	
4	Overfitting and underfitting	
5	Reproducibility issues	

Monitoring Maintenance and Lifecycle Management		
1	Lack of monitoring	
2	Model drift	
3	Retraining complexity	
4	Version management issues	

Organizational and Human Factors		
1	Change resistance	
2	Cross-functional friction	
3	Skill gaps	
4	Unclear ownership	

Problem Definition and Business Alignment		
1	Misalignment with business value	
2	Over-automation expectations	
3	Poorly defined use cases	
4	Scope creep	

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